



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Spring 22 23

Section: C

Software Quality Assurance and Testing

Parental Supervision Application

A Report submitted

By

SN	Student Name	Signature	Student ID
1	JANNATUL TABBASSUM HIRA		20-43255-1
2	LAMEYA RAHMAN		19-39795-1
3	SUMAITA ISLAM NAJA		19-41267-3
4	SYEDA NAZIA TASNIM		20-41997-1

Checked By Industry Personnel

Name:

Designation:

Company:

Sign:

Date:

Software Test Plan

for

Parental Supervision Application

Version 1.0 approved

Prepared by JANNATUL TABBASSUM HIRA (20-43255-1), LAMEYA RAHMAN (19-39795-1), SUMAITA ISLAM NAJA (19-41267-3), SYEDA NAZIA TASNIM (20-41997-1)

Department of Computer Science (CS)

American International University-Bangladesh (AIUB)

26/04/2023

Checked By Industry Personnel

Name:

Designation:

Company:

Sign:

Date:

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Revision History:

<i>Revision</i>	<i>Date</i>	<i>Updated by</i>	<i>Comments</i>
1	15/04/2023	JANNATUL TABBASSUM HIRA	First Draft
2	18/04/2023	LAMEYA RAHMAN	Second Draft
3	20/04/2023	SUMAITA ISLAM NAJA	Third Draft
4	24/04/2023	SYEDA NAZIA TASNIM	Fourth Draft
5	26/04/2023	JANNATUL TABBASSUM HIRA	Fifth Draft

1 TEST PLAN IDENTIFIER: RS-MTP_ParentalSupervisionApplication_01.3

2 REFERENCES

1. Figma: <https://www.figma.com/file/nj7o7IxEpZTZ139oe6X7J5/Untitled?node-id=0-1&t=0tNdyAShFwwZ4GN3-0>
2. Selenium: www.selenium.com
3. Monday.com: <https://monday.com/>

3 INTRODUCTION

3.1 Background to the Problem

Every child should take care of their aging parents, even if they are busy. We need to create a society that cares for those who are in need, especially our elderly citizens. Younger generations should learn to value their parents and make it a priority to care for them as they grow older. However, some people get too caught up in their careers and families that they forget their responsibilities. Regardless of how busy you are, children have a duty to look after their parents.

3.2 Solution to the Problem

To make caring for our elderly parents easier, we have come up with a device called "Parental Supervision". It has features that help maintain their daily needs, safety, and health.

Parents and their children will have separate accounts with their voter ID information. Each parent's account will have their daily routine/needs and medical reports. Users will receive notifications about their parent's important daily needs and medications. Parents can let their children know when they have taken their medications by pressing a button. If they forget, their children will be alerted and receive emergency alerts to help their parents.

4 REQUEIREMNT SPECIFICATION:

4.1 System Features

- Registration
- Create Profile
- Schedule
- Notification
- Update & Information
- Settings & background options

4.1.1 Registration

Functional Requirements

- The program should allow child users to register by providing their personal information such as name, email address, and country.
- A verification code should be sent to their email for verification. If the code is not received within 60 seconds, the system should have a "send code again" button for the user to request another code.

Priority Level: HIGH

Preconditions: The user should have a valid email address and password.

4.1.2 Create Profile

Functional Requirements

- After signing in, the child user should be able to create two profiles - one for themselves and one for their guardian.
- Both profiles should require personal information such as name, phone number, age, and location. Once the profiles are created, they should be linked together.

Priority Level: HIGH

Preconditions: The user should be signed in and have completed the registration process.

4.1.3 Schedule

Functional Requirements

- After creating the profiles, the child user should be able to schedule their guardian's activities such as mealtime, medication, workout time, and other important information.

Priority Level: HIGH

Precondition: The user should have completed the profile creation process.

4.1.4 Notification

Functional Requirements

- The software should send notifications to the guardian's device to remind them of their scheduled activities.
- If the guardian does not respond, a notification should be sent to the child's profile to inform them.

Priority Level: HIGH

Precondition: The user should have scheduled activities

4.1.5 Update & Information

Functional Requirements

- The user should be able to update their schedule and personal information such as email address and phone number.

Priority Level: HIGH

Precondition: The user should have completed the registration process and scheduled activities.

4.1.6 Settings & Background Options

Functional Requirements

- The software should allow users to change the language and choose between dark mode and light mode.

Priority Level: LOW

Precondition: The user should have completed the registration process and scheduled activities.

4.2 System Quality Attributes

Non-Functional Attributes

Usability

The system should be easy to use for older people, so it needs to be simple and user-friendly.

Priority Level: HIGH

Flexibility

Users should be able to customize the system to fit their needs easily. The software should be organized based on user requests, and users should be able to make changes easily.

Priority Level: HIGH

Integrity

Any changes made to the system should be securely stored, and the system should be able to track all user actions securely.

Priority Level: HIGH

Efficiency

The system needs to be resource-efficient to provide good performance. Since the program is primarily for older people, it should be designed to be small and efficient.

Priority Level: HIGH

Performance

The system's performance depends on how well it's optimized. The system should have high-quality components for database servers, transactions, and usage capacity to keep the system running smoothly.

Priority Level: **HIGH**

Reusability

The system should be designed to be reusable to save costs. The program should use different code library classes that can be easily reused in different application modules.

Priority Level: **HIGH**

Portability

The system should be easy to transfer from one medium to another.

Priority Level: **MEDIUM**

Testability

The system should be easy to test, and any errors should be easy to find. The system should be divided into different modules for testing.

Priority Level: **HIGH**

Maintainability

The system should be well-documented and designed for easy maintenance. The system should not need to be shut down for maintenance more than once every 24 hours. Maintenance should be cost-effective and easy.

Priority Level: **HIGH**

Interface

The system should have a simple point-and-click interface with text fields, buttons, and other graphical components. The interface should be consistent and designed to accommodate users with disabilities like color blindness. The system should also provide clear and understandable messages for each step and activity.

Priority Level: **HIGH**

4.3 System Interface

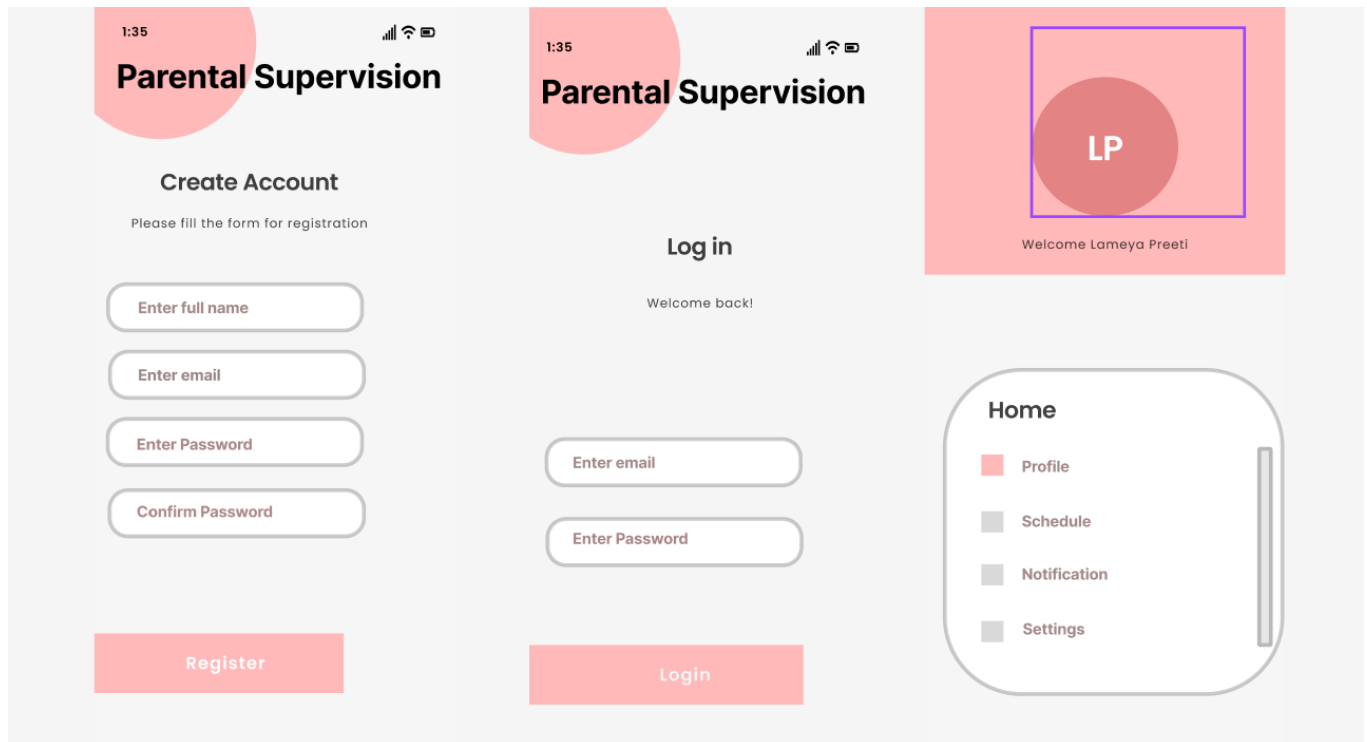


Figure1: Parental Supervision App UI/UX

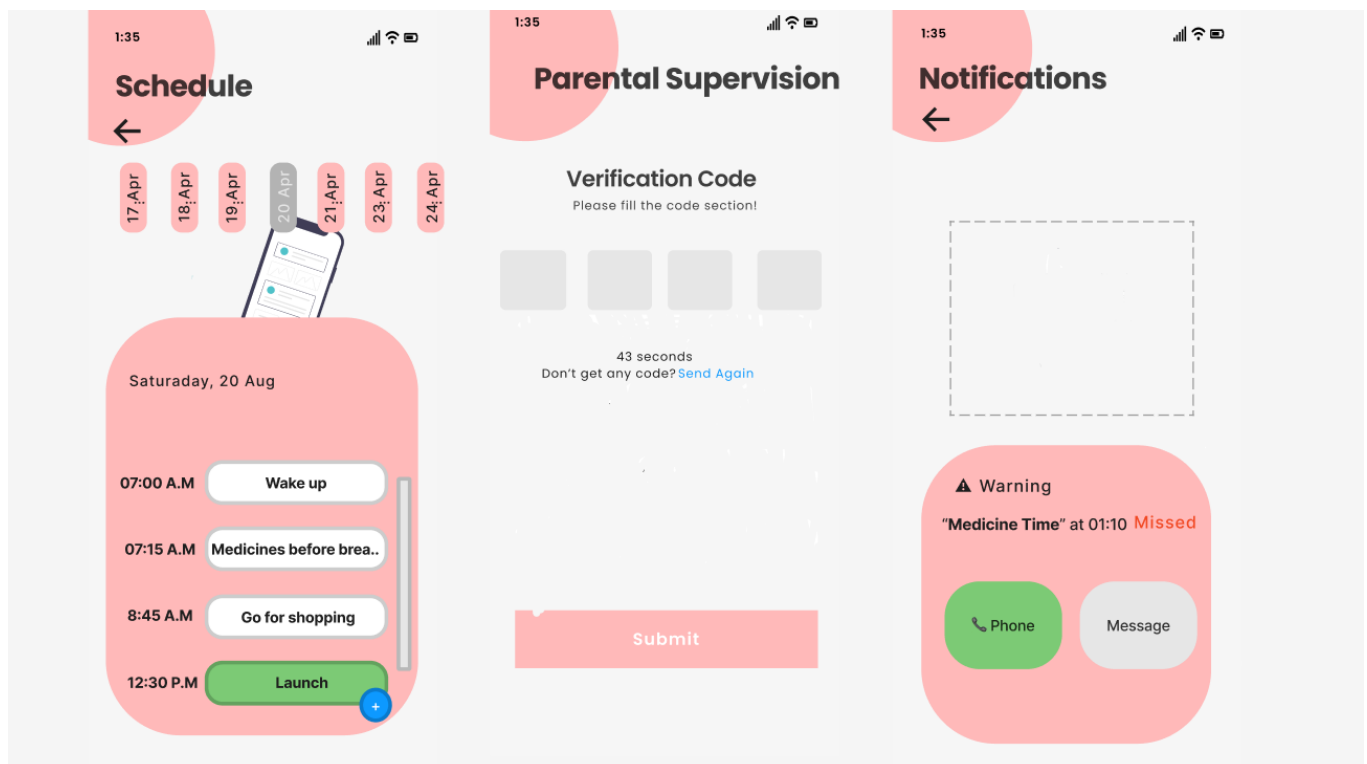


Figure2: Parental Supervision App UI/UX

4.4 Project Requirements

1. 3 system developers
2. 2 software testers
3. 4 high-end custom PC
4. 2 Android & IOS mobile phones
5. LAN connection
6. Other
7. Budget

Budget Details:

Total Development Time: 80days.

Total working Hours required for developing: $(12\text{hr} * 80\text{days}) = 960$ hours

960Hours (12hr per day)

Total working Hours required for testing $(8\text{hr} * 20\text{days}) = 160$ hours for Testing & Debugging

60 days for Revision

Project Developers: 270,000TK (each developer)

3 developers will get: $(3 * 270,000) = 810,000\text{TK}$

Project Testers: 175,000TK (each tester)

2 testers will get: $(2 * 175,000) = 350,000\text{TK}$

Cost of PC's: 90,000TK (each PC) = $(4 * 90,000) = 360,000\text{TK}$

2 smartphones = 95,000TK (each phone) = $(2 * 95,000) = 190,000\text{TK}$

LAN connection: 117,000TK

Subscriptions: 17,475TK

Other: 55,000TK

Total Budget: $(810,000 + 350,000 + 360,000 + 55,000 + 190,000 + 117,000 + 17,475)$ TK

= 1,899,475TK

5 FEATURES NOT TO BE TESTED

Our project has 10 features, but we won't be testing 2 of them - Profile Edit and Multi-Language.

Profile Edit: Lets users change their profile information, like their picture and app theme. However, since we'll already check their vital information during registration, this feature isn't necessary for the app to work properly.

Multi-Language: Allows users to choose their preferred language for the app. We did a survey and found that most users prefer English. We might add the option to use Bangla later, but for now, it's not a high priority. Therefore, we won't test this feature right now.

6 TESTING APPROACH

6.1 Testing Levels

- The testing of a "Parental Supervision Application" will go through different levels of testing, including Unit, Framework, Integration, and Acknowledgment levels. However, due to budget and plan constraints, most of the testing will be done by the test supervisor with the help of the development teams.
- Unit Testing involves testing individual parts of the application to ensure they work properly. This includes both manual testing and testing the code itself. For example, when a developer finish designing a page in the application, they will test it to make sure it works as intended.
- Control Flow Testing looks for mistakes in the code caused by improper programming. This is done by the developer choosing a specific part of the code to test and creating a Control Flow Chart to represent the testing process.
- Data Flow Testing is used to identify program inconsistencies by tracking the data used by the program. The developer will look for abnormalities in the data, such as data that is defined but not used, or used but not defined. They will then try to fix any issues they find.
- Domain Testing checks whether the software takes inputs within the allowable range and generates the correct output. The developer will identify domains from the source code and analyze the interpreted predicates to determine the domains.
- Integration Testing checks how well different parts of the application work together. This is done by assembling the different modules and checking for any issues or conflicts.
- System Testing checks the overall functionality of the system from the customer's perspective. This is done by testing the entire system, including the hardware and software.
- Acceptance Testing is done by the customer or end-user to ensure that the system meets their requirements.

6.2 Test Tools

When building software, we need to test it to make sure it works properly. There are different types of tests we can do.

- One type is called unit testing. This means testing each small part of the software separately to make sure it works correctly.
- We use a tool called Selenium to do unit testing for software .
- Another type of testing is integration testing. This means testing how different parts of the software work together to make sure they work properly.
- We also use Selenium for integration testing, but for web-based software.
- System testing is another type of testing. This is when we test the whole software to make sure it works as it should. We use Selenium for system testing too.

Selenium is a tool that helps us automate the testing process, making it faster and more efficient.

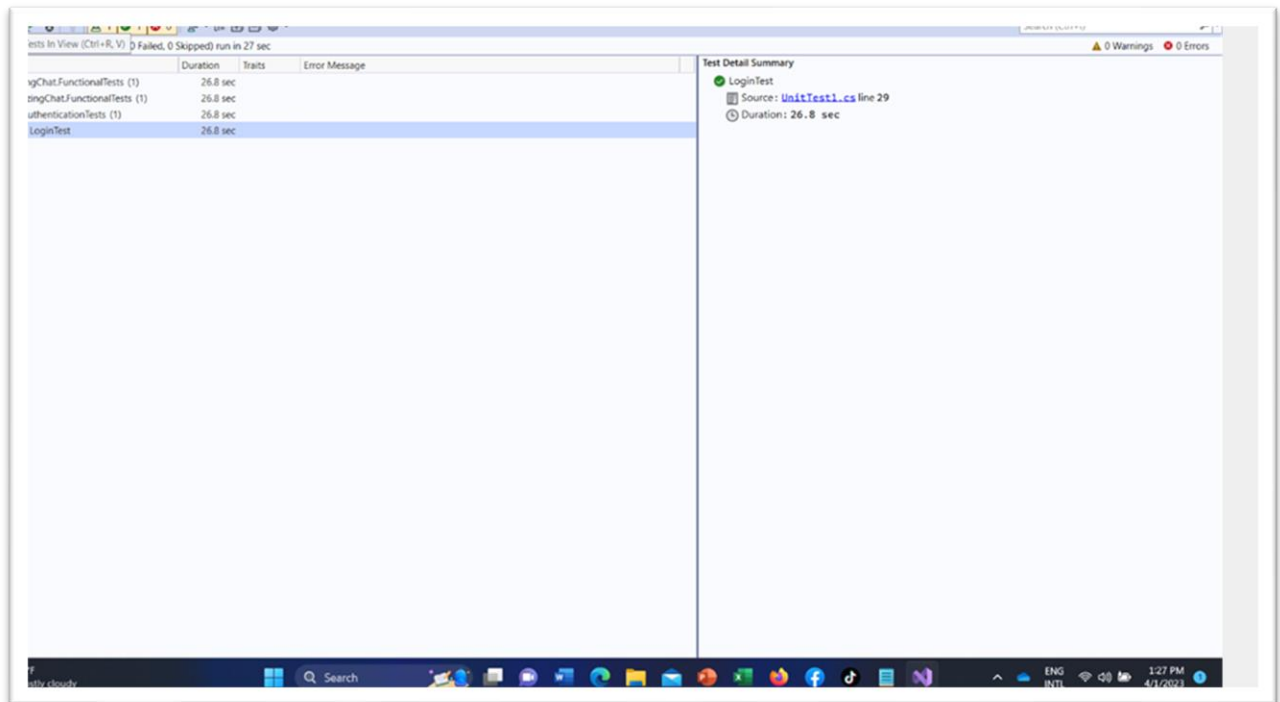


Figure1: Selenium Testing demo

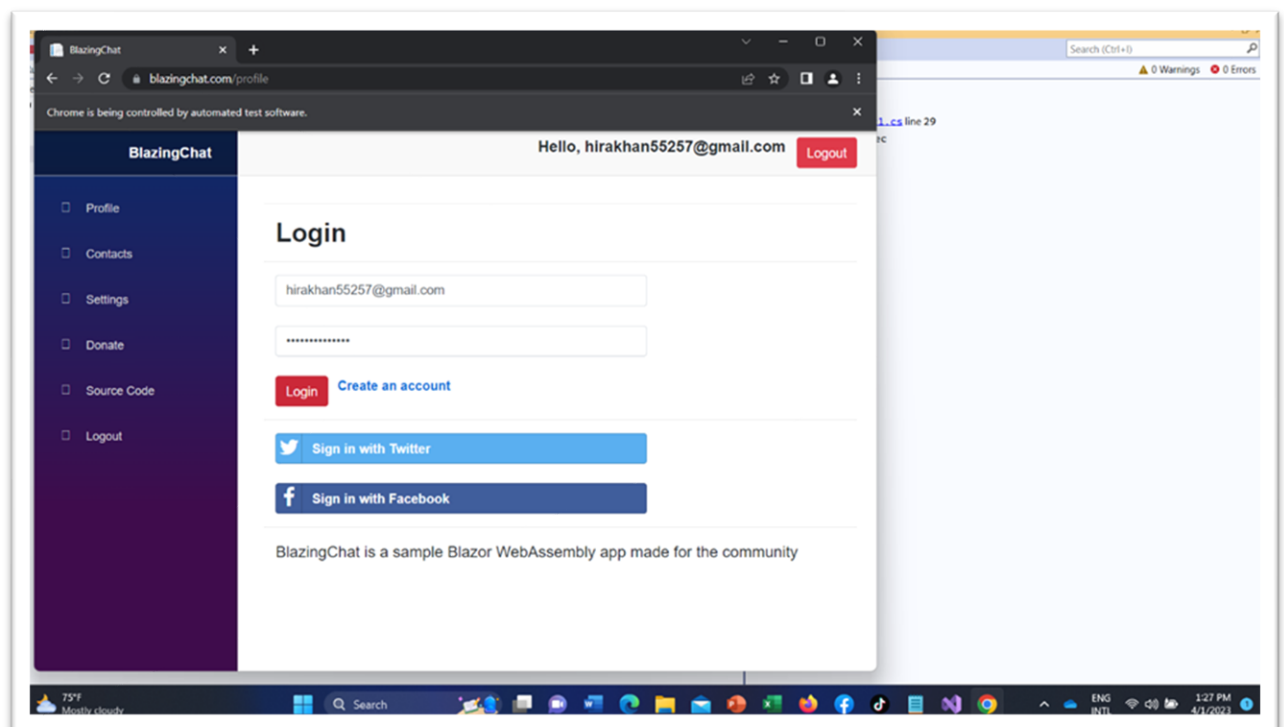


Figure2: Selenium Testing demo

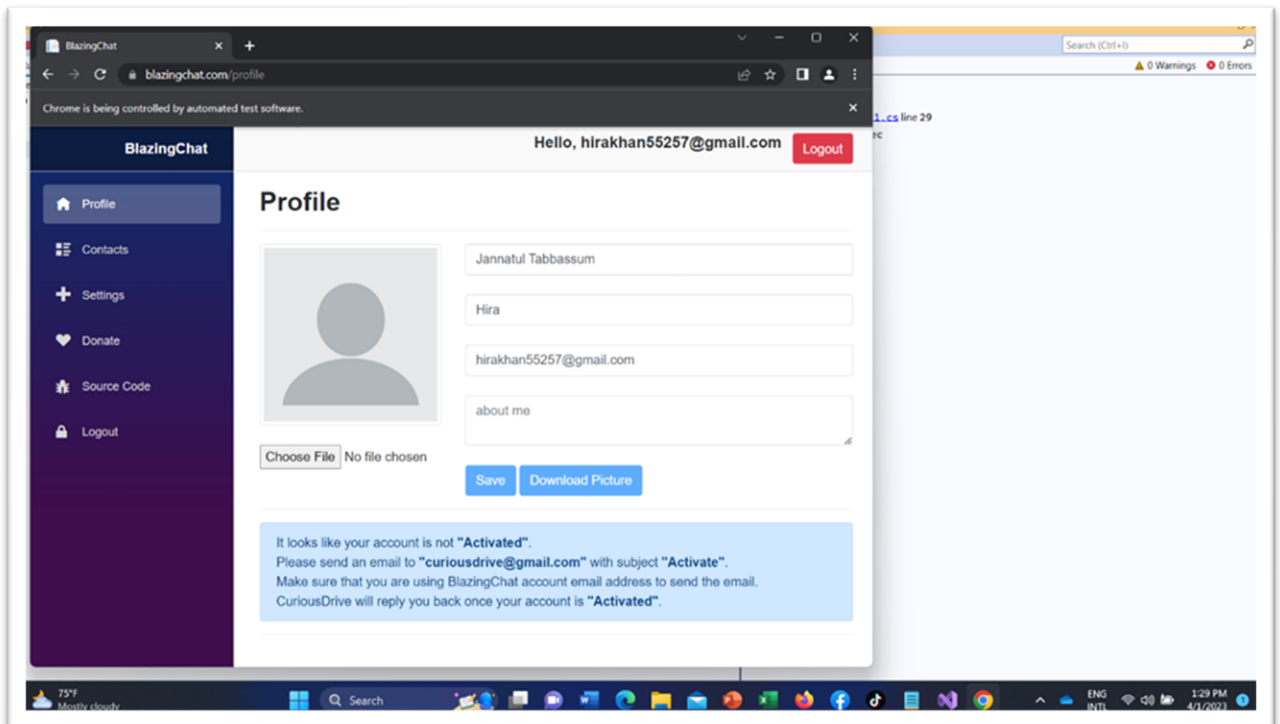


Figure3: Selenium Testing demo

6.3 Meetings

The project lasted for 80 days, with the testing phase taking 20 days. During this time, all modules went through unit testing, integration testing, and system testing in the first four weeks. The QA Engineers were responsible for the entire testing process and if they found any issues, they would fix them and retest the whole module. The following two weeks were dedicated to client or third-party testing, which included acceptance testing. The system was tested, and an assessment report was provided. If any problems were identified, the review report was retrieved and inspected to correct the issue. It took a week to complete the necessary changes and finalize the examination. Every Friday at 10:00 PM, a general meeting was held.

7 TEST CASES/TEST ITEMS

Project Name: Parental Supervision	Test Designed by: JNNATUL TABBASSUM HIRA
Test Case ID: AN0001	Test Designed date: 25/03/23
Test Priority: High	Test Executed by: JNNATUL TABBASSUM HIRA

Module Name: Registration			Test Execution date: 26/03/23	
Test Title: Registration Verification				
Description: Security code and verify the email id.				
Precondition (If any): 1. If the email id is valid.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Verify Id by code	Email id: Name: Rabby	Give confirmation	As expected,	pass
Post Condition: The system will allow for further step.				
Project Name: Parental Supervision			Test Designed by: JNNATUL TABBASSUM HIRA	
Test Case ID: AR00200			Test Designed date: 26/3/23	
Test Priority: High			Test Executed by: JNNATUL TABBASSUM HIRA	
Module Name: Schedule			Test Execution date: 27/3/23	
Test Title: Check the selected schedule				
Description: Verify the users routine.				
Precondition (If any): Schedule must be organized.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. See all schedule 2. Select preferable routine	Schedule: Morning medicine	Should see the schedule	As expected,	pass
Post Condition: The system will allow for further step.				
Project Name: Parental Supervision	Test Designed by: JNNATUL TABBASSUM HIRA			
Test Case ID: AR00201	Test Designed date: 27/3/23			
Test Priority: High	Test Executed by: JNNATUL TABBASSUM HIRA			
Module Name: Notification Alarm	Test Execution date: 28/3/23			
Test Title: Verification of Timely Notification				
Description: Check that does the notification follow the routine.				
Precondition (If any): Routine time must be organized.				
Test Steps	Test Data	Expected Results	Actual Results	Status
				(Pass/Fail)
1. Verify Child profile notification 2. Verify Parent Profile notification	Schedule Data: 23289240923	Notification alarm must give alerts.	As expected,	Pass
Post Condition: Show the notification				

Project Name: Parental Supervision			Test Designed by: Nazia	
Test Case ID: AR00202			Test Designed date: 28 / 3/23	
Test Priority: High			Test Executed by: Nazia	
Module Name: Time Format			Test Execution date: 29/3/23	
Test Title: set the time format accordingly				
Description: Verify the users routine.				
Precondition (If any): format is set via AM-PM method				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
3. See the frame of schedule 4. Select format accordingly	Schedule: For Morning medicine use: AM	Should see the schedule format	As expected,	pass
Post Condition: The system will allow for further step.				

Project Name Parental Supervision	Test Designed by: Nazia
Test Case ID: AR00203	Test Designed date: 30 / 3/23
Test Priority: High	Test Executed by: Nazia
Module Name: App installation	Test Execution date: 2/4/23
Test Title: Run the app on the expected device	
Description: To use the system application make sure to download it properly	
Precondition (If any): Download the application from Play store	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Find the application in the play store 2.press install to download it	Installation: Search by the name of the application for downloading: parental security application	Should see the expected app icon	As expected,	pass
Post Condition: The system will allow for further step.				

Project Name: Parental Supervision			Test Designed by: Nazia	
Test Case ID: AR00204			Test Designed date: 3 / 4/23	
Test Priority: High			Test Executed by: Nazia	
Module Name:			Test Execution date: 5/4/23	
Test Title: set the time format accordingly				
Description: Verify the users routine.				
Precondition (If any): format is set via AM-PM method				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
5. See the frame of schedule 6. Select format accordingly	Schedule: For Morning medicine use: AM	Should see the schedule format	As expected,	pass
Post Condition: The system will allow for further step.				

		Test Designed by: Nazia		
Project Name: Parental Supervision				
Test Case ID: AR00205		Test Designed date: 6 /4/23		
Test Priority: High		Test Executed by: Nazia		
Module Name: Notification Alarm		Test Execution date: 7 / 4 /23		
Test Title: Verification of Timely Notification				
Description: Check that if the notification pop up followed by the routine.				
Precondition (If any): Routine time must be organized.				
Test Steps	Test Data	Expected Results	Actual Results	Status
				(Pass/Fail)
1.Verify Child profile notification 2. Verify Parent Profile notification	Schedule Data: 23289240923	Notification alarmmust give alerts.	As expected,	Pass
Post Condition: Show the notification				

Project Name: Parental Supervision	Test Designed by: Nazia
Test Case ID: AR00206	Test Designed date: 8 / 4/23
Test Priority: High	Test Executed by: Nazia
Module Name: Get the schedule Alert	Test Execution date: 9/4/23

Test Title: Set alarm to get reminder				
Description: To get alerts for medicine routine you have to set an alarm for reminder				
Precondition (If any): Set the time according the schedule				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Put the exact time in the schedule box. 2.add label for the time so that receiver can understand	Time: 2.30 Label: morning medicine (Syrup-50mg)	Should get the alarm with label	As expected,	pass
Post Condition: The system will allow for further step.				

Project Name: Parental Supervision		Test Designed by: JNNATUL TABBASSUM HIRA		
Test Case ID: AR00207		Test Designed date: 10 / 4/23		
Test Priority: High		Test Executed by: JNNATUL TABBASSUM HIRA		
Module Name: login Session		Test Execution date: 13/4/23		
Test Title: Verify login with valid username and password				
Description: Test the application login				
Precondition (If any): use valid name and password				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1.Go to application main interface		Enter valid username and password		
2.Enter a valid user name and password	Username: hira@gmail.com		As expected,	pass
3.click submit for login	Password: 12345678			
Post Condition: The system will allow for further step.				

8 ITEM PASS/FAIL CRITERIA

There are 3 ways to decide if something is good or bad.

- First, a group of people will check it. This group includes the UAT Manager, UAT Analyst, Project Sponsor, and a member of the Advancement Team.
- Next, there are some requirements that need to be met. These include finishing all the code, passing all the integration tests, and fixing any major issues.
- Finally, there is a 4-step process to decide if the thing is good or bad. First, they summarize the results of the testing. Then they look at how it affects the company. Finally, they decide if it's good enough or not. If it's good, they accept it. If it's not good enough, they may accept it with some conditions or reject it completely.

9 TEST DELIVERABLES

For our Parental Supervision Application App, we need to create some important documents and reports to make sure everything works properly.

- First, we need to create a plan for acceptance tests, which will help us make sure the app meets all the requirements.
- We also need a plan for system and integration tests to check if everything works together.
- And, we need plans for unit tests, which help us test individual parts of the app.
- Next, we need to create screen prototypes and report mock-ups to show what the app will look like. This will help us make sure that everything is user-friendly and easy to understand.
- During testing, we need to keep track of any defects or incidents that come up and create reports summarizing them.
- We also need to keep test logs and create turnover reports to share with others who will be working on the app.

All of these deliverables are important for making sure our Parental Supervision Application App works well and is easy to use.

10 STAFFING AND TRAINING NEEDS

- To ensure success in using mobile applications, it's important to provide training to the developers and analysts about the basics of the application.
- The sales team should also receive training on the new features and reporting.
- Additionally, at least one member of the development and operations team should be trained on how to set up and manage the mobile-based wholesaler system.

11 RESPONSIBILITIES

Name	Role	Responsibilities
JANNATUL TABBASSUM HIRA	Project Manager	1. Determines which of the identified problems must be solved and in what order. 2. Review all open reports each week, identifying the problems of interaction Obtain the necessary literature to find deficient information about any problem.
LAMEYA RAHMAN & SUMAITA ISLAM NAJA	Test Lead	1. Defining the testing activities, All responsibilities of test planning 2. To check if the term has all the necessary resources to execute the testing activities. 3. Prepare the report of testing activities. Updating project managers regularly about the progress of testing activities.

SYEDA NAZIA TASNIM & JANNATUL TABBASSUM HIRA	Test Engineer	1. Develop test cases and prioritize testing activities. 2. Execute all the test cases and report defects. 3. Inform the test lead about what all resources will be required for software testing.
SUMAITA ISLAM NAJA & LAMEYA RAHMAN	Software Tester	1. Designing testing scenarios for testing. 2. Perform the testing. Create testing documentation and does testing-related work.
JANNATUL TABBASSUM HIRA & SYEDA NAZIA TASNIM	Software Tester	1. Creation of test designs, test processes, test cases, and test data. 2. Carry out testing as per the defined procedures 3. Prepare all reports related to software testing carried out.

12 TESTING SCHEDULE

We have added the following tasks to our project plan. We used a cloud-based tool called Monday.com to create a Gantt chart that shows when each task will take place. The plan includes the exact dates and times for each activity. The project manager will work with the leaders of the development and testing teams to figure out which staff members are needed for each task. This includes the testing team, development team, management, and the client.

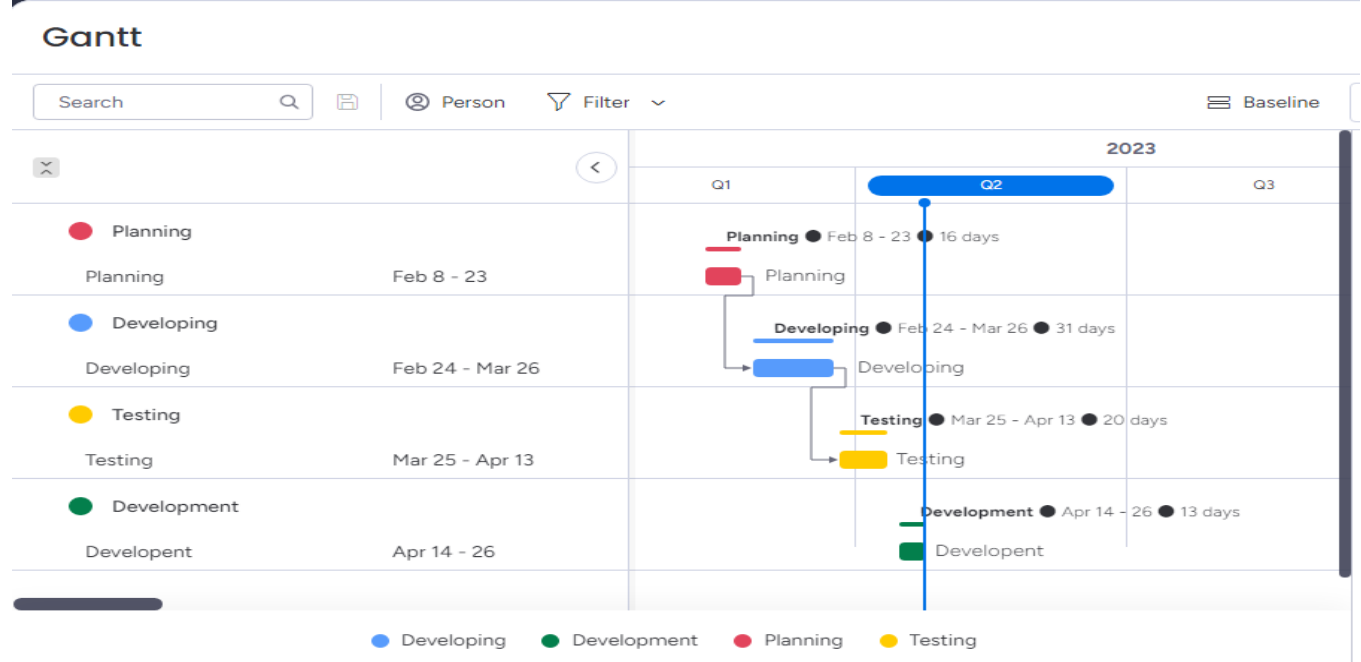


Figure6: Gantt chart for project schedule.

13 PLANNING RISKS AND CONTINGENCIES

Risk	Probability	Impact	Mitigation
Error in Functions	Medium	Medium	Test the application frequently and maintain a daily backup.

Wrong SQL Command for Important Data	Medium	High	Maintain security checks & backups.
Loss of encrypted data	Medium	High	Maintain security check and backup.
User's account hacking attempt	High	High	Restrict user after three unsuccessful login attempts in an hour.

14 APROVALS:

Position	Assigned
Project Sponsor	JANNATUL TABBASSUM HIRA
Development Management	LAMEYA RAHMAN SUMAITA ISLAM NAJA
EDI Project Manager	JANNATUL TABBASSUM HIRA SUMAITA ISLAM NAJA
RS Test Manager	JANNATUL TABBASSUM HIRA SYEDA NAZIA TASNIM
RS Development Team Manager	JANNATUL TABBASSUM HIRA SYEDA NAZIA TASNIM

Reassigned Sales	SUMAITA ISLAM NAJA LAMEYA RAHMAN
Order Entry EDI Team Manager	SYEDA NAZIA TASNIM LAMEYA RAHMAN